







INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR, ADYAR - CHENNAI

ZcannerReality:AI-Based Universal Virtual Try On And Digital Twin Generation

PROJECT REPORT

Submitted by

Jaison J(24619143)

Gnaneshwaran S(24619112)

Karthikeyan P(24619147)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

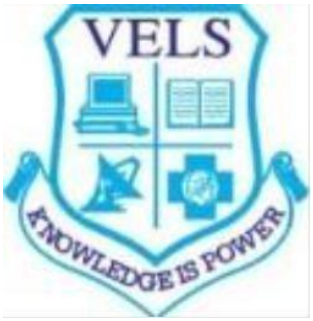
SCHOOL OF ENGINEERING

VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED STUDIES

(VISTAS)

CHENNAI

OCTOBER 2026



in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY CSE –

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

BONAFIDE CERTIFICATE

Certified that this project report titled "ZcannerReality: AI-Based Universal Virtual Try-On and Digital Twin Generation" is the bonafide work of Jaison J (24619143), Gnaneshwaran S (24619112), and Karthikeyan P (24619147), who carried out the project work under my supervision. Certified further that, to the best of my knowledge, the work reported here does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE OF THE HOD
Dr. K. Ulaga Priya, M.E., Ph.D.
Associate Professor & Head
Department of AI & ML
School of Engineering
VISTAS
Chennai – 600 117

SIGNATURE OF THE SUPERVISOR
Dr. K. Ulaga Priya, M.E., Ph.D.
Associate Professor & Head
Department of AI & ML
School of Engineering
VISTAS
Chennai – 600 117

Submitted for the Project Work and Viva-Voce examination held on …… at Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai – 600 117.

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

We hereby declare that the project work entitled “ZcannerReality: AI-Based Universal Virtual Try-On and Digital Twin Generation”, submitted to [Name of the Institution], in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning), is a record of original project work carried out by us under the supervision of Dr. Ulaga Priya K. We further declare that this report has not been submitted previously, in part or in full, for the award of any degree, diploma, associateship, or fellowship at this or any other university or institution.

Jaison J (24619143)

Gnaneshwaran S (24619112)

Karthikeyan P (24619147)

ACKNOWLEDGEMENT

It is certainly a great delight and proud privilege to acknowledge the invaluable support, guidance, and encouragement received from numerous individuals throughout the development and completion of this project.

First and foremost, we express our profound gratitude and heartfelt thanks to our respected FOUNDER & CHAIRMAN, Shri. Dr. ISHARI K. GANESH, for providing excellent infrastructural facilities and a stimulating academic environment that fostered our technical growth and learning.

We convey our sincere thanks and deep gratitude to our respected HEAD OF THE DEPARTMENT, Dr. K. Ulagapriya, for her continuous support, constructive feedback, administrative assistance, and encouragement throughout all phases of this project.

We owe our deepest gratitude and warmest thanks to our esteemed PROJECT SUPERVISOR AND GUIDE, Dr. Ulaga Priya K, for her invaluable guidance, insightful suggestions, constructive reviews, and unwavering encouragement throughout the conceptualization, system design, prototype implementation, and preparation of this project report.

We also extend our sincere appreciation to the Project Coordinator, faculty members, technical staff, and laboratory staff of the Department of Computer Science and Engineering, for their constant assistance, technical advice, and support, which helped us overcome various challenges during the development of our project, "ZcannerReality: AI-Based Universal Virtual Try-On and Digital Twin Generation."

Last but not least, we express our heartfelt love and gratitude to our beloved parents and friends for their unconditional moral support, patience, and constant motivation, without which this endeavour would not have been possible.

Jaison J (24619143)

Gnaneshwaran S (24619112)

Karthikeyan P (24619147)

TABLE OF CONTENTS

ABSTRACT 7

LIST OF ABBREVIATIONS 8

CHAPTER 1 – INTRODUCTION 10
1.1 General Introduction 10
1.2 Background and Motivation 10
1.3 Problem Description 11
1.4 Existing System 11
1.5 Limitations of the Existing System 12
1.6 Proposed System 13
1.7 Objectives of the Project 13
1.8 Scope of the Project 14
1.9 Major Contributions 15
1.10 System Modules – Overview 15
1.11 Organization of the Report 16
1.12 Chapter Summary 16

CHAPTER 2 – LITERATURE REVIEW 17
2.1 Deep Learning-Based Object Detection 17
2.2 Open-Vocabulary Object Detection 17
2.3 Virtual Try-On Technology 17
2.4 Human Parsing and Image Preprocessing 18
2.5 Web-Based AI Applications and System Integration 18
2.6 Research Gap and Motivation 18
2.7 Existing Approaches, Limitations, and Research Gap 19
2.8 Chapter Summary 20

CHAPTER 3 – SYSTEM DESIGN AND ARCHITECTURE 21
3.1 System Overview 21
3.2 Existing-System Architecture 22
3.3 Proposed-System Architecture 22
3.4 Overall System Workflow 24
3.5 Module 1: Footwear Detection and Classification 25
3.6 Module 2: Image Processing and Human Parsing 26
3.7 Module 3: Virtual Try-On Processing 26
3.8 System Output and Result Interpretation 27
3.9 User Interaction and Result Presentation 27
3.10 Hardware Requirements 28
3.11 Software Requirements 28
3.12 Chapter Summary 28

CHAPTER 4 – IMPLEMENTATION AND METHODOLOGY 29
4.1 Development Environment 29
4.2 Hardware and Execution Environment 30
4.3 Dataset Preparation 30
4.3.1 Footwear Detection Dataset 31
4.3.2 VITON-HD Dataset 31
4.3.3 Image Preprocessing 31

4.4 Footwear Detection Model Implementation 32
4.5 Virtual Try-On Model Integration 33
4.6 Image Processing and Human Parsing 33
4.7 Streamlit Application Development 34
4.8 Testing and Validation Methodology 34
4.9 Model Inference and Result Visualization 35
4.10 Implementation Challenges and Future Enhancements 35
4.11 Chapter Summary 35

CHAPTER 5 – RESULTS AND DISCUSSION 36
5.1 Experimental Setup 36
5.2 Performance of the Footwear Detection Module 37
5.3 Performance of the Virtual Try-On Module 37
5.4 Integration into the Streamlit Application 38
5.5 Footwear Detection and Virtual Try-On: Functional Comparison 38
5.6 Overall Analysis 39
5.7 Quantitative Evaluation Metrics 40
5.8 Limitations and Threats to Validity 40
5.9 Chapter Summary 40

CHAPTER 6 – ZCANNERREALITY APPLICATIONS AND DEMONSTRATION 41
6.1 Application Overview 42
6.2 Live Camera and Virtual Try-On Interface 42
6.3 Live Camera and Try-On Output Demonstration 43
6.4 System Status and Performance Information 44
6.5 Potential Applications 44
6.6 Chapter Summary 44

CHAPTER 7 – CONCLUSION AND FUTURE WORK 45
7.1 Conclusion 45
7.2 Achievement of Project Objectives 46
7.3 Main Contributions 46
7.4 Limitations 46
7.5 Future Enhancements 47
7.6 Final Summary 47

CHAPTER 8 – REFERENCES 48

CHAPTER 9 – APPENDIX 49
9.1 Prototype Artefacts 49
9.2 Information to Be Completed by the Authors 49

CHAPTER 10 – SOURCE CODE DOCUMENTATION 50
10.1 Overview of the Implementation Codebase 51
10.2 Footwear Detection Using YOLO 51
10.3 Dataset Configuration and Model Training 52
10.4 Virtual Try-On Processing and Application Interface 53

CHAPTER 11 – CONFERENCE APPLICATION 55
Conference Publication Details 56

ABSTRACT

Online shoppers often face challenges when selecting footwear and clothing, as product images do not always provide a clear understanding of how items will look when viewed in a real-world context. Traditional online shopping platforms offer limited opportunities for personalized product visualization and virtual try-on. ZcannerReality is an AI-based virtual try-on and digital twin generation system designed to enhance the online shopping experience by combining computer vision, deep learning, and virtual try-on technology. The system incorporates a footwear detection module using YOLOv8 and YOLO-World to identify and classify footwear into categories such as boots, heels, sandals, shoes, slippers, and sneakers. A web-based interface developed using Streamlit provides users with an accessible platform to interact with the system. The virtual try-on component integrates a clothing try-on model and image-processing techniques to visualize garments on a user-provided person image. Computer vision techniques, including human parsing and mask generation, are used to prepare input images for virtual try-on processing. The current prototype focuses on footwear detection and virtual try-on integration, while broader digital twin capabilities and support for multiple product categories remain areas for future development. The project aims to provide a more interactive and personalized shopping experience, helping customers visualize products before making purchasing decisions.

Keywords — Virtual try-on, digital twin, YOLOv8, YOLO-World, computer vision, deep learning, Streamlit, human parsing, image processing, e-commerce.

LIST OF FIGURE LIST OF ABBREVIATIONS

Abbreviation	Full Form / Definition
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
GPU	Graphics Processing Unit
IoT	Internet of Things
ML	Machine Learning
OpenCV	Open Source Computer Vision Library
RGB	Red, Green, Blue
SegFormer	Semantic Segmentation Transformer
UI	User Interface
URL	Uniform Resource Locator
VITON-HD	High-Resolution Virtual Try-On Dataset and Method
YOLO	You Only Look Once
YOLOv8	You Only Look Once Version 8
YOLO-World	Open-Vocabulary Object Detection Model
AI	Artificial Intelligence
CNN	Convolutional Neural Network

VELS
INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR, ADYAR - CHENNAI



ZCANNERREALITY

AI-Based Universal Virtual Try-On and Digital Twin Generation

Scan. Try Virtually. Shop Confidently.

ABSTRACT

ZcannerReality is an AI-based virtual try-on and digital twin generation system designed to enhance the online shopping experience by combining computer vision, deep learning, and virtual try-on technology. The system uses YOLOv8 and YOLO-World for footwear detection and classification, and integrates a virtual try-on model to visualize garments on a user image. A web-based interface developed using Streamlit provides an easy-to-use platform for product detection, item selection, and virtual try-on. The current prototype focuses on footwear detection and virtual try-on integration, while broader digital twin capabilities and support for multiple product categories remain areas for future development.

OBJECTIVES

1. Detect and classify footwear items using YOLOv8 / YOLO-World.
2. Provide an interactive web interface using Streamlit.
3. Integrate a virtual try-on model for clothing and footwear.
4. Use human parsing and mask generation for try-on processing.
5. Enable users to visualize products on a person image.
6. Lay the foundation for digital twin generation and multi-category try-on in future.

METHODOLOGY

1. INPUT

- User image (upload / camera)
- Product image (footwear / clothing)

2. PRODUCT DETECTION

- YOLOv8 / YOLO-World
- Detect footwear items
- Classify category (boots, heels, sandals, shoes, slippers, sneakers, etc.)
- Localize bounding box

3. HUMAN PARSING & PREPROCESSING

- Human parsing (SegFormer)
- Generate garment masks
- Align person and product
- Preprocess images

4. VIRTUAL TRY-ON MODEL

- Clothing try-on model (e.g., VITON-HD based)
- Footwear try-on integration
- Image synthesis and blending

5. OUTPUT

- Try-on result image
- Detected product details
- Web interface output (Streamlit)

DATASETS USED

Dataset	Purpose	Details
Footwear Detection Dataset	Train and test footwear detection model	Footwear images with categories (boots, heels, sandals, shoes, slippers, sneakers, etc.)
VITON-HD	Virtual try-on of clothing	High-resolution dataset with person images, clothing items and annotations

IMPLEMENTATION TOOLS

Tool / Library	Purpose
Python	Programming language
YOLOv8 / YOLO-World	Footwear detection and classification
SegFormer	Human parsing and mask generation
Virtual Try-On Model	Clothing and footwear try-on
OpenCV	Image processing
Streamlit	Web-based user interface
PyTorch	Model training and inference

RESULTS & DISCUSSION

Footwear Detection Output



Detected footwear with bounding box and class label

Detected Product Details

Class	Confidence
Sneaker	0.92
Shoe	0.85
Sandal	0.81
Boot	0.78

Virtual Try-On Output



Person image, product image and try-on result

Web Interface (Streamlit)



PERFORMANCE COMPARISON

Model / Method	Detection Accuracy (%)	Try-On Quality	Inference Time (s)
YOLOv8	-	-	-
YOLO-World	-	-	-
Virtual Try-On Model (VITON-HD based)	-	-	-
ZcannerReality (Ours)	-	-	-

KEY OBSERVATIONS

- Footwear items are detected and classified using YOLOv8 / YOLO-World.
- Virtual try-on model generates try-on results for user-provided images.
- Streamlit interface provides an interactive and easy-to-use platform.
- The current prototype focuses on footwear detection and virtual try-on integration.
- Support for multiple product categories and digital twin generation is planned for future work.

CONCLUSION

ZcannerReality demonstrates the potential of combining computer vision and virtual try-on technology to enhance the online shopping experience. The current prototype successfully integrates footwear detection, virtual try-on, and a web-based interface. Future work will aim to extend the system to multiple product categories and digital twin generation for a more comprehensive and personalized shopping experience.

CHAPTER 1
INTRODUCTION

1.1 General Introduction

Online shopping has become an important part of modern retail, allowing customers to purchase footwear and clothing without visiting physical stores. However, one of the major challenges of online shopping is that customers cannot physically try products before purchasing them. Product images provide only a limited view of how an item may appear when selected or worn by a person. This can make it difficult for customers to evaluate the appearance of footwear and clothing, compare products, and make confident purchasing decisions. Traditional e-commerce platforms primarily depend on product images, descriptions, and size charts, offering limited opportunities for personalized visual interaction.

ZcannerReality is an AI-based virtual try-on and digital twin generation system designed to improve the online shopping experience by integrating computer vision, deep learning, image processing, and virtual try-on technology. The system incorporates a footwear detection module based on YOLOv8 and YOLO-World to identify and classify footwear into categories such as boots, heels, sandals, shoes, slippers, and sneakers. A web-based interface developed using Streamlit allows users to interact with the system, upload images, and access its available functionalities. The virtual try-on component uses a clothing try-on model and image-processing techniques, including human parsing and mask generation, to prepare images for visualization.

The current prototype focuses on footwear detection and virtual try-on integration. The broader objective is to develop a more comprehensive digital shopping environment in which customers can visualize products on their own images before making purchasing decisions. Advanced digital twin generation, expanded product-category support, and more comprehensive product visualization are considered future development areas.

1.2 Background and Motivation

In conventional online shopping, customers generally rely on static product photographs, descriptions, and size information to decide whether a product meets their preferences. Although these methods provide useful information, they cannot fully represent how a product may look when worn by a particular person. Footwear selection can be particularly challenging because customers may want to compare different styles, such as sneakers, sandals, boots, and heels, before choosing a product. Similarly, clothing customers may want to understand how a garment could appear on their own image rather than viewing it only on a model.

Recent developments in artificial intelligence and computer vision provide opportunities to address these limitations. Object detection models such as YOLOv8 and YOLO-World can identify and classify visual objects, while virtual try-on techniques can generate visual representations of clothing on a person image. Human parsing and mask generation help identify relevant body regions and prepare images for the try-on process. By integrating these technologies through a user-friendly web interface, ZcannerReality aims to make product exploration more interactive and personalized.

The motivation behind ZcannerReality is to reduce the gap between viewing a product online and understanding its potential appearance when worn. The system combines footwear detection with virtual try-on functionality to provide a foundation for more engaging digital shopping

experiences. Its Streamlit interface offers an accessible way to interact with the available modules without requiring users to work directly with complex machine-learning tools.

ZcannerReality is intended to support customers in visualizing and exploring products, rather than guarantee their actual fit, comfort, or physical appearance. The current prototype represents an initial step towards a more comprehensive virtual shopping platform. Future enhancements may include try-on support for additional product categories, improved image alignment, real-time camera interaction, and more advanced digital twin generation.

1.3 Problem Description

In online shopping, customers often find it difficult to determine how footwear and clothing will look when worn by them. Traditional e-commerce platforms mainly provide product photographs, descriptions, and size charts, which may not be sufficient for customers to visualize products on their own images. As a result, customers may struggle to compare different styles, understand the visual appearance of products, and make confident purchasing decisions. Although computer vision and virtual try-on technologies offer potential solutions, integrating product detection, image processing, and virtual visualization into a single accessible platform remains a technical challenge.

ZcannerReality addresses the following core problems:

1. Limited Product Visualization: Traditional online shopping platforms primarily display static product images, making it difficult for customers to visualize how footwear or clothing may appear when worn.

2. Difficulty in Product Identification and Classification: Identifying and categorizing footwear from images can be challenging when products have different shapes, designs, and visual characteristics. An automated object detection system is required to identify footwear and classify it into relevant categories.

3. Lack of Personalized Virtual Try-On: Customers generally cannot visualize selected products directly on their own images. Virtual try-on technology is needed to generate a visual representation of a selected product on a person image.

4. Image Processing and Alignment Challenges: Virtual try-on requires appropriate image preparation, including human parsing, mask generation, and alignment of relevant image regions. Inaccurate preprocessing may affect the quality of the generated result.

5. Limited Integration and Accessibility: Product detection and virtual try-on are often implemented as separate functionalities. Integrating these capabilities through a user-friendly web interface can make AI-based product visualization more accessible to customers.

Therefore, ZcannerReality aims to integrate footwear detection, image processing, virtual try-on functionality, and a Streamlit-based interface into a unified platform for interactive and personalized online shopping.

1.4 Existing System

Existing online shopping platforms and AI-based product visualization approaches can be grouped into the following categories:

- Static Product Image-Based Shopping: Conventional e-commerce platforms display product photographs, descriptions, prices, and size charts. Customers must rely on these details to imagine how a product may look when worn.

- Manual Product Search and Selection: Customers browse product categories and filters to find suitable items. Product identification and comparison depend largely on the available platform features and user interaction.

- Object Detection-Based Systems: Computer vision models, such as YOLO-based detectors, identify and classify objects in images. These systems can support automated footwear recognition, but object detection alone does not generate a virtual try-on result.

- Virtual Try-On Systems: AI-based virtual try-on methods generate images showing how clothing may appear on a person. Their output quality depends on factors such as image quality, body-region parsing, garment masks, and image alignment.

- Image Processing and Human Parsing Systems: Image-processing techniques and human parsing models identify relevant regions of an image and prepare inputs for virtual try-on. These techniques generally require integration with other components to provide a complete user-facing application.

- Web-Based AI Applications: Frameworks such as Streamlit allow machine-learning models to be presented through interactive web interfaces. However, a web interface alone does not provide footwear detection or virtual try-on unless the corresponding models and processing pipelines are integrated.

Limitation of the Existing System: These approaches address individual aspects of online product visualization, but they do not necessarily provide an integrated workflow combining footwear detection, image preprocessing, virtual try-on, and an accessible user interface. ZcannerReality aims to bring these capabilities together in a single prototype, with broader digital twin generation and multi-category product visualization planned as future enhancements.

1.5 Limitations of the Existing System

The existing online shopping platforms and AI-based product visualization systems have several limitations that affect the shopping experience and the effectiveness of virtual product presentation. The major limitations are as follows:

1. Limited Product Visualization: Most conventional e-commerce platforms rely on static product images and descriptions. Customers cannot directly visualize how a selected product may appear on their own images before making a purchase.

2. Limited Product Detection and Classification: Traditional shopping platforms generally depend on predefined product categories, search filters, and manual selection. They may not provide integrated AI-based footwear detection and classification from user-provided images.

3. Lack of Integrated Virtual Try-On: Product detection and virtual try-on are often handled by separate systems. The absence of an integrated workflow can make product exploration less convenient and interactive.

4. Image Alignment and Processing Challenges: Virtual try-on systems may produce unsatisfactory results when the input images have poor quality, body regions are incorrectly identified, or garment masks and image alignment are inaccurate.

5. Limited Personalization: Static product displays do not fully account for differences in individual appearance and preferences. Customers may find it difficult to compare products visually in a personalized manner.

6. Computational Resource Requirements: Deep-learning-based detection and virtual try-on models may require significant computational resources, particularly when processing high-resolution images. This can limit their accessibility on systems without dedicated GPUs.

7. Restricted Product Category Support: Some virtual try-on systems focus primarily on clothing or a specific product category. Extending the same platform to support footwear, clothing, accessories, and other products requires additional models, preprocessing methods, and integration.

ZcannerReality aims to address these limitations by integrating AI-based footwear detection, image preprocessing, virtual try-on functionality, and a Streamlit-based interface. However, the quality of generated results, computational efficiency, and broader multi-category support depend on further testing and development.

1.6 Proposed System

ZcannerReality is an AI-based virtual try-on and digital twin generation system designed to improve the online shopping experience by integrating footwear detection, computer vision, image processing, and virtual try-on technology. The system uses YOLOv8 and YOLO-World to identify and classify footwear from images into categories such as boots, heels, sandals, shoes, slippers, and sneakers. A virtual try-on component is integrated to support the visualization of clothing on a person image. Human parsing and mask-generation techniques are used to prepare the required inputs for the try-on process. A web-based interface developed using Streamlit allows users to access the available functionalities through a simple and interactive platform.

The current prototype focuses on footwear detection and virtual try-on integration. The footwear detection module processes input images and identifies relevant footwear objects, while the virtual try-on pipeline uses person images, product images, and image preprocessing techniques to generate visual results. The system architecture is designed to support further integration of additional product categories and more advanced digital twin capabilities. Features such as comprehensive multi-category try-on, real-time camera-based visualization, and advanced digital twin generation are considered future development stages rather than completed functionalities.

1.7 Objectives of the Project

ZcannerReality has five major objectives, summarized in Table 1.1 along with their status in the current prototype.

Objective	Description	Status in Current Prototype
Footwear Detection and Classification	Identify footwear items in images and classify them into categories such as boots, heels, sandals, shoes, slippers, and sneakers.	Implemented using YOLOv8 / YOLO-World. Detection performance should be reported using measured evaluation results.

Virtual Try-On	Generate a visual representation of selected clothing on a person image to help users explore products before purchasing.	Virtual try-on integration is part of the prototype. Final model details and output quality should be confirmed using actual results.
Image Processing and Human Parsing	Prepare person and product images using image processing, human parsing, and mask generation for virtual try-on.	Included in the virtual try-on processing workflow; the exact completion status depends on the final integrated pipeline.
Interactive Web Interface	Provide a simple interface for uploading images, accessing footwear detection, and using virtual try-on functionality.	Streamlit-based interface implemented for the current prototype.
Digital Twin and Multi-Category Expansion	Establish a foundation for personalized product visualization and future support for clothing, footwear, accessories, and other product categories.	Broader digital twin generation and comprehensive multi-category support remain future development goals.

Table 1.1: ZcannerReality Project Objectives and Prototype Status

1.8 Scope of the Project

The scope of ZcannerReality encompasses the following areas:

- Footwear Detection: Identification and classification of footwear items from images into categories such as boots, heels, sandals, shoes, slippers, and sneakers using YOLOv8 and YOLO-World.

- Virtual Try-On: Visualization of selected clothing on a person image using an integrated virtual try-on model and image-processing techniques.

- Image Processing and Human Parsing: Preparation of person and garment images through preprocessing, human parsing, and mask generation to support the virtual try-on workflow.

- Interactive Web Interface: Providing a Streamlit-based interface through which users can upload images, access footwear detection, and interact with the available virtual try-on functionality.

- Personalized Product Visualization: Establishing a foundation for helping customers explore footwear and clothing visually before making online purchasing decisions.

- Digital Twin and Future Expansion: Extending the system towards more comprehensive digital twin generation, additional product categories, improved visualization, and real-time camera-based interaction.

- Application Domain: Supporting the development of AI-assisted e-commerce applications, virtual product demonstrations, and interactive online retail platforms.

The current project focuses on footwear detection and virtual try-on integration. The broader capabilities described under future expansion require additional implementation and evaluation.

1.9 Major Contributions

The principal contributions of the ZcannerReality project are:

1. AI-Based Footwear Detection: Integration of YOLOv8 and YOLO-World for identifying and classifying footwear items, providing an automated alternative to manual image-based identification.

2. Virtual Try-On Integration: Incorporation of a clothing virtual try-on workflow to support the visualization of garments on a person image, making product exploration more interactive.

3. Image Preprocessing Workflow: Use of image-processing techniques, including human parsing and mask generation, to prepare the input images required by the virtual try-on component.

4. Web-Based Application Interface: Development of a Streamlit-based interface that brings the available AI functionalities into an accessible application, reducing the need for users to interact directly with model code.

5. Integrated System Architecture: Establishment of a prototype architecture that combines footwear detection, virtual try-on processing, and user interaction within one project.

6. Foundation for Digital Twin Development: Design of a foundation for future enhancements involving broader product-category support, personalized product visualization, and more comprehensive digital twin generation.

These contributions establish the current prototype as a foundation for AI-assisted virtual shopping. Further development and evaluation are needed to establish the performance, scalability, and effectiveness of advanced features.

1.10 System Modules — Overview

ZcannerReality consists of four major modules that work together to provide AI-based footwear detection and virtual try-on functionality.

- Module 1 (Footwear Detection and Classification): Processes input images using YOLOv8 and YOLO-World to identify footwear objects and classify them into categories such as boots, heels, sandals, shoes, slippers, and sneakers.

- Module 2 (Image Processing and Human Parsing): Prepares person and garment images for virtual try-on by applying image preprocessing, human parsing, and mask-generation techniques. These operations help identify relevant regions required by the try-on model.

- Module 3 (Virtual Try-On Processing): Integrates a clothing virtual try-on model to generate a visual representation of selected garments on a person image. The generated output depends on the model, input images, and preprocessing quality.

- Module 4 (Web Interface and System Integration): Uses Streamlit to provide a user-friendly interface for accessing the available modules, uploading images, and viewing detection and virtual try-on outputs.

The overall architecture provides a foundation for extending the system to additional product categories and more advanced digital twin capabilities.

1.11 Organization of the Report

The remainder of this report is organized as follows:

- Chapter 1 (Introduction): Introduces the project background, problem description, existing system, proposed system, objectives, scope, major contributions, and system modules.

- Chapter 2 (Literature Review): Reviews relevant research on object detection, open-vocabulary detection, virtual try-on, human parsing, and AI-based product visualization. It also identifies the research gaps addressed by the project.

- Chapter 3 (System Analysis and Design): Describes the system requirements, proposed architecture, workflow, module interactions, and hardware and software requirements.

- Chapter 4 (Datasets and Data Preparation): Explains the footwear detection dataset, the VITON-HD dataset, image preprocessing, dataset organization, and preparation of inputs for the relevant models.

- Chapter 5 (Implementation and Methodology): Describes the implementation of footwear detection using YOLOv8 and YOLO-World, virtual try-on integration, human parsing, mask generation, and the Streamlit interface.

- Chapter 6 (Testing, Results, and Discussion): Presents the available system outputs, testing procedures, detection results, virtual try-on demonstrations, and discussion of limitations. Quantitative performance results should be included only when supported by actual evaluation.

- Chapter 7 (Applications, Limitations, and Future Scope): Discusses potential applications in e-commerce, current system limitations, and future enhancements such as additional product categories and advanced digital twin generation.

- Chapter 8 (Conclusion): Summarizes the project, its current implementation, contributions, and opportunities for further development.

- References, Appendix, and Source Code Documentation: Provide the research references, supplementary materials, implementation evidence, and relevant technical documentation.

1.12 Chapter Summary

This chapter introduced ZcannerReality as an AI-based virtual try-on and digital twin generation system intended to improve the online shopping experience. It discussed the limitations of conventional product visualization, described the proposed integration of footwear detection and virtual try-on technology, and presented the project objectives, scope, contributions, and system modules. The chapter also outlined the organization of the report.

Chapter 2 reviews the relevant literature on object detection, open-vocabulary recognition, virtual try-on, human parsing, and image-processing techniques, establishing the technical foundation for the proposed system.

CHAPTER 2
LITERATURE REVIEW

2.1 Deep Learning-Based Object Detection

Object detection is an important area of computer vision that enables systems to identify and locate objects within images and videos. Deep-learning-based object detectors have become widely used in applications such as image recognition, automated inspection, retail analytics, and product identification. The YOLO (You Only Look Once) family of models performs object detection by predicting object locations and classes within a unified framework, making it suitable for applications that require efficient image processing.

For ZcannerReality, YOLOv8 provides a foundation for detecting and classifying footwear items in images. The model can identify objects belonging to categories such as boots, heels, sandals, shoes, slippers, and sneakers when trained or configured for the relevant classes. This capability supports automated footwear recognition and can reduce dependence on manual product identification. However, detection performance depends on factors such as dataset quality, object size, image clarity, and variation in footwear appearance.

2.2 Open-Vocabulary Object Detection

Conventional object detection models generally recognize a predefined set of classes, which can limit their flexibility when identifying unfamiliar objects. Cheng et al. introduced YOLO-World, an open-vocabulary object detection approach designed to detect objects using text descriptions and visual information. This approach offers greater flexibility for applications involving diverse objects and changing product categories.

In ZcannerReality, YOLO-World is relevant because footwear products exhibit considerable variation in shape, colour, design, and style. Open-vocabulary detection provides an opportunity to explore flexible product identification beyond a fixed set of categories. When combined with an appropriate footwear dataset and evaluation process, this approach can support the development of more adaptable product detection functionality. Nevertheless, open-vocabulary capability does not automatically guarantee accurate recognition of every footwear category.

2.3 Virtual Try-On Technology

Virtual try-on technology uses computer vision and generative models to create images showing how clothing may appear on a person. Traditional online shopping usually presents garments as separate product images, requiring customers to imagine how the selected item might look when worn. Virtual try-on methods address this limitation by combining a person image with a garment image to generate a synthesized visual result.

The VITON-HD method, presented by Choi et al., focuses on high-resolution virtual try-on and addresses challenges associated with misalignment between clothing and the person image. Its approach is relevant to systems that require visually detailed garment representations. For ZcannerReality, VITON-HD provides an important research foundation for understanding garment transfer, image alignment, and the preservation of person-image details during try-on generation.

However, virtual try-on output quality depends on the model architecture, input resolution, garment representation, and preprocessing pipeline. The VITON-HD dataset and method

primarily address clothing try-on and should not be treated as a footwear detection dataset or as a complete solution for footwear try-on.

2.4 Human Parsing and Image Preprocessing

Human parsing is a computer vision task that divides a person image into meaningful semantic regions, such as the head, torso, arms, and legs. This information can help virtual try-on systems determine which parts of an image should be preserved, modified, or replaced during garment generation. Image masks provide additional guidance by defining the regions associated with the clothing and the areas that should remain unchanged.

Xie et al. introduced SegFormer, a transformer-based semantic segmentation architecture designed to combine hierarchical feature extraction with an efficient segmentation decoder. Its design makes it relevant to applications requiring semantic understanding of image regions.

In ZcannerReality, human parsing, garment masks, and clothing-agnostic masks are relevant preprocessing components for the virtual try-on workflow. Appropriate preprocessing can help prepare person and garment images for the selected try-on model. Incorrect segmentation or mask generation, however, may introduce visual artefacts or reduce the quality of the generated output.

2.5 Web-Based AI Applications and System Integration

Machine-learning models often require a user-friendly interface to make their functionality accessible to people who do not have programming experience. Web application frameworks allow model inference, image uploads, and result visualization to be integrated into a single application.

Streamlit is a Python-based framework that supports the development of interactive data applications. It is suitable for building prototypes that combine image inputs, model execution, and output presentation without requiring a complex frontend implementation.

ZcannerReality uses Streamlit as its application interface to connect the available footwear detection and virtual try-on functionalities. This integration provides a common environment for users to submit images and view processing results. The effectiveness of the application depends on the integration of the underlying models, the management of image inputs, and the computational resources available during inference.

2.6 Research Gap and Motivation

The reviewed research demonstrates the individual value of object detection, open-vocabulary recognition, virtual try-on, semantic segmentation, and web-based AI applications. However, these techniques address different aspects of visual product interaction and require suitable integration to support a complete shopping workflow.

The main research gaps motivating ZcannerReality are:

1. Limited integration of footwear detection and virtual try-on: Product recognition and clothing visualization are distinct tasks that require coordination within a common application.

2. Restricted personalization in conventional e-commerce: Static product images provide limited opportunities for customers to explore products using their own images.

3. Preprocessing challenges in virtual try-on: Human parsing, mask generation, and image alignment can influence the quality of synthesized clothing images.

4. Accessibility of AI-based product visualization: Detection and try-on models need to be presented through a straightforward interface so that users can interact with them without directly managing model code.

5. Expansion towards broader digital twin capabilities: A foundation is needed for future extensions involving additional product categories and more comprehensive personalized visualization.

ZcannerReality addresses these gaps through a prototype that combines YOLO-based footwear detection, virtual try-on processing, image preprocessing, and a Streamlit interface. Its broader digital twin capabilities and expanded multi-category support remain potential areas for future development.

References for Chapter 2

[1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[2] Cheng et al., "YOLO-World: Real-Time Open-Vocabulary Object Detection," Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. Available online.

[3] S. Choi et al., "VITON-HD: High-Resolution Virtual Try-On via Misalignment-Aware Normalization," Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021. Available online.

[4] E. Xie et al., "SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers," Advances in Neural Information Processing Systems (NeurIPS), 2021. Available online.

[5] Ultralytics, "YOLO-World Model Documentation." Available online.

[6] P. S. H. Choi et al., "VITON-HD Project and Resources." Available online.

2.7 Existing Approaches, Limitations, and Research Gap

Table 2.1 summarizes the relevant research studies and their contributions to the development of ZcannerReality.

Study / Technology	Method	Contribution	Limitation Relevant to This Work
Redmon et al. (2016)	YOLO object detection	Introduced real-time object detection using a unified deep-learning	Detection depends on the trained classes and may require adaptation for

		framework.	specialized footwear categories.
Cheng et al. (2024)	YOLO-World open-vocabulary detection	Supports object detection using visual features and text-based category descriptions.	Open-vocabulary detection may not consistently identify every footwear category without suitable prompts and evaluation.
Choi et al. (2021)	VITON-HD virtual try-on	Introduced high-resolution clothing try-on with a focus on garment and person-image misalignment.	Focuses on clothing visualization and does not directly provide footwear detection or a complete shopping application.
Xie et al. (2021)	SegFormer semantic segmentation	Provides an efficient transformer-based approach to semantic segmentation.	Segmentation alone does not generate virtual try-on images or identify footwear product categories.
Ultralytics YOLO documentation	YOLO model implementation and inference	Provides practical tools for training, prediction, and integration of YOLO-based models.	Application performance depends on model configuration, dataset quality, and available computing resources.
Streamlit documentation and framework	Python-based web application development	Supports interactive interfaces for image uploads, model inference, and output visualization.	A web framework does not independently provide object detection or virtual try-on capabilities.

Table 2.1: Summary of Reviewed Literature and Limitations

The reviewed studies demonstrate that object detection, open-vocabulary recognition, semantic segmentation, and virtual try-on can address different aspects of visual product interaction. However, each technique has a specific purpose and does not independently provide a complete personalized shopping experience.

The main research gaps addressed by ZcannerReality are:

1. Limited integration: Footwear detection and virtual try-on are different tasks that need to be integrated into a common application.

2. Limited personalized visualization: Conventional online shopping relies primarily on static product images, making it difficult for customers to visualize selected clothing on their own images.

3. Image preprocessing challenges: Virtual try-on requires suitable person images, garment masks, and human parsing to support the generation of useful visual results.

4. Accessibility: AI models require an accessible interface through which users can upload images and view the outputs without directly interacting with model code.

5. Limited product-category coverage: Extending product visualization to footwear, clothing, accessories, and other categories requires additional model integration and preprocessing methods.

The research gap addressed by ZcannerReality is therefore the integration of AI-based footwear detection, virtual try-on processing, image preprocessing, and a web-based interface within a unified prototype. The project establishes a foundation for more personalized product visualization, while comprehensive digital twin generation and broader multi-category support remain future development goals.

2.8 Chapter Summary

This chapter reviewed the research foundations relevant to ZcannerReality, including YOLO-based object detection, open-vocabulary detection, high-resolution virtual try-on, semantic segmentation, and web-based AI application development. The literature review identified key limitations in conventional product visualization, independent AI modules, image preprocessing, and personalized virtual try-on.

Based on these findings, ZcannerReality proposes an integrated prototype combining footwear detection, virtual try-on functionality, and a Streamlit-based interface. The next chapter presents the system analysis, requirements, architecture, and design of the proposed system.

CHAPTER 3
SYSTEM DESIGN AND ARCHITECTURE

3.1 System Overview

ZcannerReality follows an integrated workflow that combines image acquisition, footwear detection, image preprocessing, virtual try-on, and result visualization. The system is designed to improve the online shopping experience by allowing users to identify footwear products and visualize clothing through AI-based image processing techniques.

The workflow begins when a user uploads an image through the Streamlit interface. Based on the selected functionality, the image is processed by the footwear detection module or passed to the virtual try-on pipeline. The footwear detection module uses YOLOv8 and YOLO-World to identify and classify footwear items. The virtual try-on module processes person and garment images using image preprocessing, human parsing, and mask-generation techniques. The prepared images are then passed to the selected virtual try-on model to generate a visual output.

Finally, the system displays the detection results or generated virtual try-on image through the web interface. The current prototype focuses on footwear detection and virtual try-on integration, while advanced digital twin generation and broader product-category support remain future development goals.

3.2 Existing-System Architecture

Conventional online shopping platforms primarily depend on static product photographs, product descriptions, specifications, and size charts. Customers browse product catalogues, compare available items, and make purchasing decisions based on the information provided.

However, this approach has several limitations. Static images do not allow customers to visualize clothing directly on their own images. Product identification generally depends on manual browsing, search functions, and category filters. Furthermore, footwear detection and virtual try-on are often separate functionalities rather than components of a unified application.

The conventional system follows a simple workflow:

- The customer opens an online shopping platform.
- The customer browses product categories and views product images.
- The customer compares product descriptions, prices, and sizes.
- The customer estimates how the product may look when worn.
- The customer makes a purchasing decision based on the available information.

The primary limitation of the existing system is the lack of integrated AI-based footwear detection and personalized virtual try-on. ZcannerReality aims to address this limitation by combining object detection, image processing, virtual try-on, and a user-friendly web interface.

3.3 Proposed-System Architecture

The overall monitoring workflow of VisionGuard AI is shown in Figure 3.1.

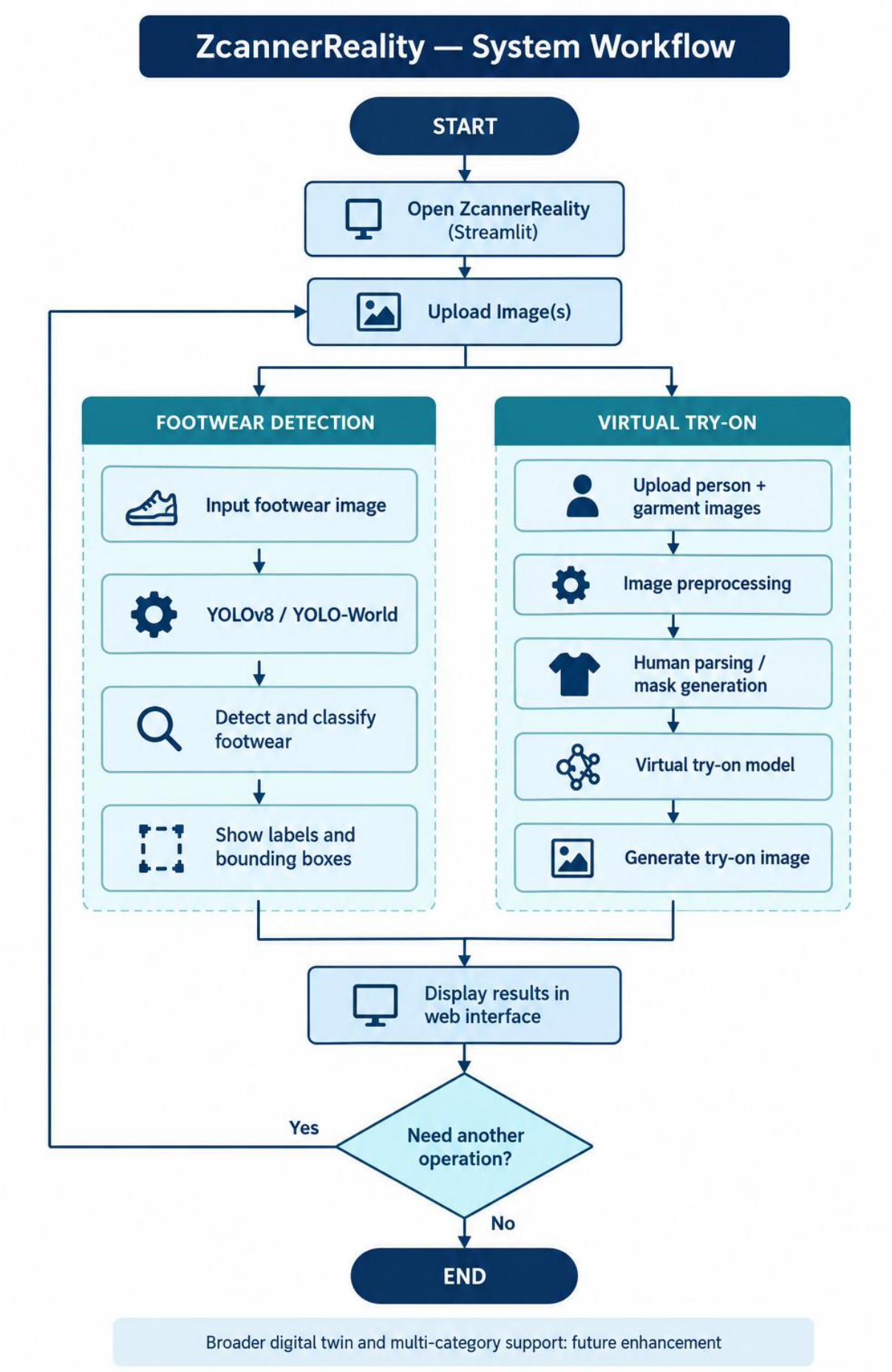


Figure 3.1: Flow Diagram Illustrating the Overall ZcannerReality Pipeline

Figure 3.2 illustrates the proposed system architecture of ZcannerReality. The workflow begins with the user accessing the Streamlit application and uploading the required images. The system then follows one of two processing pathways. In the footwear detection pathway, the uploaded footwear image is processed using YOLOv8 or YOLO-World to identify and classify footwear items. In the virtual try-on pathway, person and garment images undergo image preprocessing and human parsing, including mask generation where required, before being passed to the virtual try-on model. The outputs from both pathways are directed to the result-processing stage, where the detection information or generated try-on image is prepared for display. Finally, the results are presented through the Streamlit interface for the user to review. This modular architecture integrates footwear detection and virtual try-on within a single application while providing a foundation for future enhancements, including broader product-category support and more comprehensive digital twin generation.

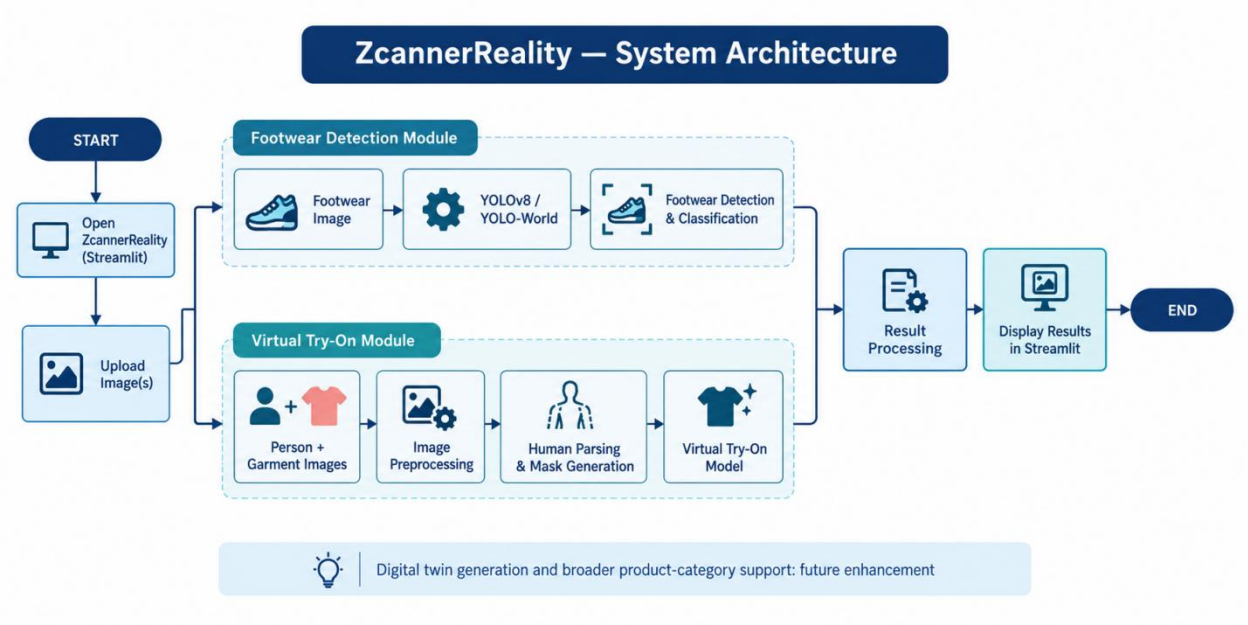


Figure 3.2: System Architecture Supporting the Proposed Zcanner Reality

3.4 Overall System Workflow

The end-to-end workflow of ZcannerReality proceeds through a series of stages, beginning with image input and ending with the visualization of detection or virtual try-on results.

1. Image Acquisition: The user opens the Streamlit application and uploads the image or images required for the selected operation.

2. Function Selection: The user selects footwear detection or virtual try-on, depending on the functionality available in the prototype.

3. Footwear Detection: The footwear image is processed using YOLOv8 or YOLO-World to identify and classify footwear objects.

4. Image Preprocessing: For virtual try-on, the person and garment images are prepared using suitable image-processing techniques. Human parsing and mask generation are used where required by the selected pipeline.

5. Virtual Try-On Processing: The prepared images are passed to the integrated clothing try-on model to generate a visualization of the garment on the person image.

6. Result Processing: The system prepares the detection information or generated image for presentation.

7. Output Display: The results are displayed through the Streamlit interface, allowing the user to examine detected footwear or view the generated try-on image.

8. Further Interaction: The user can perform another operation by providing new inputs or selecting another available functionality.

This workflow integrates the major processing stages within a single application. More advanced digital twin generation and comprehensive multi-category try-on remain future development objectives.

3.5 Module 1: Footwear Detection and Classification

Module 1 identifies and classifies footwear items from input images using computer vision and deep-learning techniques. Footwear detection is an important component of ZcannerReality because it enables the system to recognize products without requiring users to identify every item manually.

The module uses YOLOv8 and YOLO-World to process images and detect footwear objects. The detection pipeline identifies the locations of relevant objects and assigns class labels based on the model's predictions. The supported footwear categories include boots, heels, sandals, shoes, slippers, and sneakers, according to the selected dataset and model configuration.

The module follows these processing stages:

Receive a footwear image from the user interface.

Preprocess the image into the format required by the selected model.

Perform object detection and classification.

Obtain the predicted class labels and bounding-box coordinates.

Display the detected objects and associated information through the interface.

YOLOv8 provides a foundation for detecting the predefined footwear categories, while YOLO-World offers an open-vocabulary detection approach that can be explored for more flexible object recognition.

The quality of detection depends on the model configuration, dataset quality, image resolution, and visual similarity between footwear categories. Quantitative performance should be established through systematic testing rather than inferred from individual successful detections.

3.6 Module 2: Image Processing and Human Parsing

Module 2 prepares person and garment images for the virtual try-on pipeline. Image preprocessing is important because the quality and suitability of input images can influence the final generated result.

The module handles the preparation of input images and supports human parsing and mask generation where required by the selected virtual try-on model. Human parsing identifies meaningful regions of a person image, while masks help specify relevant areas for garment transfer and image synthesis.

The module includes the following processing stages:

Receive the person image and selected garment image.

Validate the input images and prepare them for processing.

Apply the required resizing and image preprocessing operations.

Generate the required garment or clothing-agnostic masks.

Prepare the processed inputs for the virtual try-on model.

SegFormer is a relevant semantic segmentation architecture for human parsing tasks. However, the exact segmentation model and preprocessing operations used by the final implementation must correspond to the integrated pipeline.

The main objective of this module is to prepare suitable inputs for virtual try-on while reducing errors caused by unsuitable image formats, incomplete masks, or incorrect image alignment.

3.7 Module 3: Virtual Try-On Processing

Module 3 generates a visual representation of selected clothing on a person image. It is intended to help users explore products visually before making purchasing decisions.

The module receives the person image, garment image, and the prepared inputs from the image-processing stage. These inputs are passed to the selected virtual try-on model, which generates a synthesized image representing the garment on the person.

The virtual try-on pipeline consists of the following stages:

Load the prepared person and garment images.

Supply the required masks and auxiliary inputs.

Execute inference using the integrated virtual try-on model.

Generate the resulting try-on image.

Return the output to the Streamlit interface for visualization.

VITON-HD provides a research foundation for high-resolution clothing virtual try-on. Its associated dataset is intended for clothing try-on rather than footwear detection. Consequently, clothing try-on and footwear detection are treated as distinct processing tasks within ZcannerReality.

The quality of the generated image depends on the selected model, input-image quality, preprocessing, mask accuracy, and available computational resources. Virtual try-on provides a visual approximation and does not guarantee actual garment fit, size, or appearance.

The current project focuses on virtual try-on integration alongside footwear detection. Extending the same functionality to comprehensive footwear try-on and additional product categories requires further model development and evaluation.

3.8 System Output and Result Interpretation

ZcannerReality produces outputs according to the selected operation. The footwear detection module returns information about detected objects, while the virtual try-on module produces a synthesized image.

The principal outputs are summarized in Table 3.1.

Output Type	Description
Detected Footwear	Identifies footwear objects present in the input image.
Class Labels	Displays the predicted footwear category for each detection.
Bounding Boxes	Indicates the detected object's location within the image.
Detection Confidence	Provides the model's confidence score when available in the output.
Virtual Try-On Image	Displays the generated clothing visualization based on the provided person and garment images.
Web Interface Output	Presents the results through the Streamlit application.

Table 3.1: ZcannerReality System Outputs

The outputs are intended to support product exploration and visual comparison. Detection confidence should not be interpreted as a guarantee of correctness, and generated try-on images should be considered visual approximations rather than exact representations of real-world fit.

3.9 User Interaction and Result Presentation

The Streamlit interface serves as the main interaction layer of ZcannerReality. It allows users to access the implemented functions without directly executing model code or managing the underlying processing pipeline.

The current architecture focuses on presenting model outputs through a common web interface. Additional functions, such as product catalogues, shopping-cart integration, automated recommendations, and real-time camera-based visualization, may be considered for future development.

3.10 Hardware Requirements

Table 3.2 lists the hardware requirements relevant to developing and running the ZcannerReality prototype.

Hardware Component	Role in the System

Laptop or Desktop Computer	Runs the application, image-processing pipeline, and AI models.
AMD Ryzen 5 5500U Processor	Provides CPU-based computation for development and inference.
32 GB RAM	Supports the application, development environment, and model execution.
Storage Device	Stores datasets, model weights, source code, and generated outputs.
Display and Input Devices	Allow developers and users to interact with the application.
Optional GPU	Can accelerate compatible deep-learning workloads when supported by the model and software environment.
Camera or Image Source	Provides person and product images when required; uploaded images can also be used.

Table 3.2: Hardware Requirements for ZcannerReality

The prototype is designed to run on a conventional laptop, with CPU-based execution considered when dedicated GPU acceleration is unavailable. Actual inference speed and memory requirements depend on the model architecture, image resolution, and selected processing pipeline.

3.11 Software Requirements

Table 3.3 summarizes the principal software technologies relevant to the project.

Software Component	Purpose
Python	Main programming language for application development and model integration.
PyTorch	Supports deep-learning model execution and compatible model development.
YOLOv8	Footwear object detection and classification.
YOLO-World	Open-vocabulary object detection and flexible object recognition.
OpenCV	Image loading, image processing, and computer vision operations.
SegFormer	Relevant semantic segmentation architecture for human parsing, where integrated.
Virtual Try-On Model	Generates clothing visualization on a person image.
Streamlit	Provides the interactive web interface.
NumPy	Supports numerical operations and image-data processing.
Pillow	Supports image loading, conversion, and manipulation where required.
Visual Studio Code	Development environment for writing and debugging the application.

Table 3.3: Software Technology Stack

The exact dependencies depend on the selected virtual try-on repository and the models integrated into the final application. Individual components may require specific Python versions,

library versions, or model weights. These requirements should be documented using the actual project environment configuration.

3.12 Chapter Summary

This chapter described the system design and architecture of ZcannerReality, including the overall workflow, footwear detection, image processing and human parsing, virtual try-on processing, system outputs, user interaction, and hardware and software requirements.

The proposed architecture integrates AI-based footwear detection and virtual try-on functionality through a Streamlit interface. It establishes a foundation for personalized product visualization while recognizing the technical challenges associated with image preprocessing, model integration, and computational requirements.

The next chapter describes the datasets, data preparation procedures, implementation environment, and methodology used to develop the ZcannerReality prototype.

CHAPTER 4
IMPLEMENTATION AND METHODOLOGY

4.1 Development Environment

ZcannerReality was developed using Python as the primary programming language for implementing the computer vision modules, integrating deep-learning models, and developing the web application. Visual Studio Code was used as the development environment for writing, testing, and debugging the source code.

The project uses YOLOv8 and YOLO-World for footwear detection and classification. PyTorch supports compatible deep-learning models, while OpenCV and other image-processing libraries assist in preparing and processing images. Streamlit provides the web interface through which users can upload images and access the implemented functionalities.

The principal software components and their purposes are summarized in Table 4.1.

Software / Library	Purpose
Python	Core programming language for system implementation.
Visual Studio Code	Source-code development and debugging.
PyTorch	Deep-learning model execution and compatible model integration.
YOLOv8	Footwear object detection and classification.
YOLO-World	Open-vocabulary object detection.
OpenCV	Image processing and computer vision operations.
SegFormer	Semantic segmentation and human parsing, where integrated.
Streamlit	Development of the interactive web application.
NumPy	Numerical computation and image-data manipulation.
Pillow	Image loading, conversion, and processing.

Table 4.1: Development Environment and Software Components

The exact software dependencies and versions depend on the selected detection and virtual try-on implementations. They should be documented using the environment configuration employed in the final prototype.

4.2 Hardware and Execution Environment

The ZcannerReality prototype is designed to run on a conventional laptop or desktop computer. The development environment includes an AMD Ryzen 5 5500U processor and 32 GB of RAM. These resources support application development, image processing, and CPU-based execution of compatible models.

The system accepts images through the Streamlit interface, processes them using the selected AI modules, and presents the resulting outputs. Dedicated GPU acceleration is optional for compatible workloads, although the inference time and memory requirements may vary according to model complexity and image resolution.

The principal hardware requirements are listed in Table 4.2.

Hardware Component	Purpose
AMD Ryzen 5 5500U processor	Executes application code and compatible CPU-based model inference.
32 GB RAM	Supports model loading, image processing, and application execution.
Storage device	Stores source code, datasets, model weights, and generated images.
Display and input devices	Support development and interaction with the application.
Optional compatible GPU	Accelerates supported deep-learning workloads.

Table 4.2: Hardware Requirements for the Prototype

The project does not require physiological sensors or patient-monitoring hardware because its application domain is virtual shopping and product visualization.

4.3 Dataset Preparation

Dataset preparation is an important stage in developing an object detection system because the quality, consistency, and category distribution of the data influence model performance. ZcannerReality uses a footwear detection dataset for identifying footwear objects and the VITON-HD dataset as a research resource for clothing virtual try-on.

4.3.1 Footwear Detection Dataset

The footwear detection dataset contains images of footwear belonging to multiple product categories. The categories include boots, heels, sandals, shoes, slippers, and sneakers. The dataset is used to support the training, validation, and testing of the footwear detection model.

The dataset is organized into separate subsets for training, validation, and testing. The corresponding image annotations provide the object classes and bounding-box coordinates required by the YOLO detection pipeline.

The preparation process involves the following stages:

- Obtain the footwear images and associated annotations.
- Organize the images into training, validation, and testing subsets.
- Verify the annotation format and class identifiers.
- Configure the dataset paths and category names in the data configuration file.
- Load the dataset through the selected YOLO implementation.
- Inspect sample images and annotations before model training or inference.

The footwear dataset supports the detection module, while the quality of its annotations and the distribution of its categories influence the reliability of the resulting model.

4.3.2 VITON-HD Dataset

VITON-HD is associated with high-resolution clothing virtual try-on. It provides a research foundation for understanding how a garment can be visually transferred onto a person image while addressing alignment challenges.

Within ZcannerReality, VITON-HD is relevant to the clothing virtual try-on component rather than the footwear detection module. Clothing images, person images, and associated preprocessing information must be prepared according to the requirements of the selected try-on implementation.

The dataset should not be treated as a footwear detection dataset, and its use does not automatically provide footwear try-on functionality.

4.3.3 Image Preprocessing

Image preprocessing prepares the input images for the selected detection and virtual try-on models. The specific operations depend on the model requirements.

For footwear detection, preprocessing may include image resizing, format conversion, and normalization performed by the selected inference pipeline.

For virtual try-on, the processing workflow may include:

- Loading the person and garment images.
- Resizing images to the model's required dimensions.
- Performing human parsing when required.
- Generating garment masks and clothing-agnostic masks.
- Preparing the processed images for virtual try-on inference.

The preprocessing pipeline must follow the selected model's expected input format. Incorrect image dimensions, missing masks, or inconsistent image alignment can affect the generated output.

4.4 Footwear Detection Model Implementation

The footwear detection module uses YOLOv8 and YOLO-World to identify footwear objects in images. These models provide the foundation for automated footwear recognition within the ZcannerReality application.

YOLOv8 supports object detection using trained category labels, while YOLO-World provides an open-vocabulary detection approach that can use text descriptions to identify relevant objects.

The implementation workflow consists of the following stages:

- Load the selected detection model and its configured weights.
- Receive an input image from the user interface.
- Preprocess the image according to the model's requirements.
- Execute inference to identify footwear objects.
- Retrieve the predicted class labels, bounding boxes, and confidence values where available.
- Render the detection results and return them to the interface.

The detection module supports the identification of footwear categories represented by the selected model and dataset. The availability of a particular category depends on the model's training configuration or the prompts supplied to an open-vocabulary detector.

The use of YOLOv8 and YOLO-World provides flexibility for experimenting with predefined footwear categories and text-guided detection. However, the models must be evaluated using a consistent test dataset before their relative performance can be established.

4.5 Virtual Try-On Model Integration

The virtual try-on module is designed to generate a visual representation of selected clothing on a person image. Unlike footwear object detection, which identifies objects in an image, virtual try-on requires image synthesis and suitable person and garment inputs.

The general processing workflow consists of the following stages:

- Load the person image and the selected garment image.
- Prepare the images according to the selected model's input requirements.
- Apply the required human parsing and mask-generation procedures.
- Pass the prepared inputs to the integrated virtual try-on model.
- Execute model inference to generate the try-on result.
- Display the generated image through the Streamlit interface.

The selected virtual try-on implementation determines the required model weights, preprocessing operations, image dimensions, and dependencies. VITON-HD provides a relevant research

foundation for clothing try-on, but the final output depends on the model actually integrated into the prototype.

The virtual try-on module is intended to support clothing visualization. Extending the same processing approach to footwear requires additional techniques because footwear placement and appearance involve different image regions and geometric constraints.

4.6 Image Processing and Human Parsing

Image processing and human parsing support the preparation of inputs for virtual try-on. Human parsing identifies meaningful regions in a person image, while masks define areas that the model should modify or preserve.

SegFormer is a semantic segmentation architecture relevant to this task. Where integrated, it can support the identification of human body regions needed by the preprocessing pipeline.

The image-processing module may include resizing, format conversion, mask generation, and alignment operations. The exact operations depend on the selected virtual try-on model.

The objective is to prepare consistent inputs for inference and reduce errors associated with incorrectly identified body regions or unsuitable masks. The final quality of the try-on image remains dependent on the combined performance of the preprocessing and image-synthesis stages.

4.7 Streamlit Application Development

Streamlit is used to provide a web-based interface for interacting with the ZcannerReality prototype. It allows users to access the available AI functions without directly executing the underlying Python model code.

The interface is designed around the following operations:

- Open the ZcannerReality application.
- Select the required functionality.
- Upload the input image or images.
- Initiate footwear detection or virtual try-on.
- View the resulting detection information or generated image.

The interface acts as the integration layer between the user and the underlying processing modules. It also provides a common location for displaying outputs and interacting with the implemented functionalities.

The overall user experience depends on the responsiveness of the application, model inference time, image size, and availability of the required model dependencies.

4.8 Testing and Validation Methodology

Testing is necessary to verify that the individual components operate as intended and that their integration supports the expected workflow. ZcannerReality should be evaluated at both the module level and the application level.

The proposed testing methodology includes the following activities:

- Input Validation: Verify that the application accepts supported image formats and handles missing or invalid inputs appropriately.

- Footwear Detection Testing: Check whether the model identifies footwear objects and returns the expected class labels and bounding boxes.

- Virtual Try-On Testing: Verify that person and garment images are processed and that the selected model generates an output.

- Interface Testing: Confirm that the Streamlit interface allows users to select functions, upload images, initiate processing, and view results.

- Integration Testing: Verify that the detection and virtual try-on modules interact correctly with the interface.

- Error Handling: Test how the application responds to missing model weights, incompatible inputs, or processing failures.

For footwear detection, suitable quantitative evaluation metrics include precision, recall, mean average precision (mAP), and inference time. For virtual try-on, evaluation may include visual inspection and suitable image-quality measures, depending on the availability of reference images and the chosen evaluation protocol.

Numerical performance results should be reported only after the relevant tests have been executed and documented.

4.9 Model Inference and Result Visualization

During inference, the application receives an input image, passes it to the selected processing module, and returns the resulting output to the user interface.

For footwear detection, the output may include bounding boxes, class labels, and confidence scores. For virtual try-on, the output is a generated image based on the person image, garment image, and preprocessing inputs.

The Streamlit interface displays these results so that users can inspect the detected products or view the virtual try-on visualization. The application is designed to support repeated operations, allowing users to process different images without manually executing the model pipeline each time.

The actual inference speed depends on the selected model, image resolution, available hardware, and model initialization overhead. Performance measurements should therefore be collected using a consistent execution environment.

4.10 Implementation Challenges and Future Enhancements

The integration of object detection and virtual try-on involves several technical challenges.

First, footwear detection depends on the quality of the training data and the ability of the model to distinguish visually similar product categories. Second, virtual try-on requires suitable person and garment images, accurate preprocessing, and compatibility between the input data and the selected model. Third, deep-learning inference may be computationally demanding on CPU-based systems, particularly for high-resolution image generation.

Future enhancements may include:

- Extending product visualization to additional clothing and accessory categories.
- Improving image alignment and the quality of generated try-on results.
- Adding real-time camera-based interaction.
- Developing more comprehensive personalized digital twin capabilities.
- Improving application responsiveness and model-inference efficiency.
- Integrating additional product comparison and recommendation features.

These enhancements can extend the current prototype into a more comprehensive AI-assisted shopping platform.

4.11 Chapter Summary

This chapter described the development environment, hardware requirements, datasets, image preprocessing, footwear detection implementation, virtual try-on integration, human parsing, and Streamlit application development. It also outlined the testing methodology, inference workflow, and challenges associated with model integration.

The implementation combines footwear detection and virtual try-on within a common application architecture. Further evaluation is required to establish quantitative model performance, while broader product-category support and comprehensive digital twin generation remain future development opportunities.

Chapter 5 presents the testing results and discussion of the ZcannerReality prototype, using actual outputs and measured performance where available.

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Experimental Setup

The ZcannerReality prototype was developed to integrate AI-based footwear detection, virtual try-on processing, and a Streamlit-based user interface. The system was designed to accept user-provided images, process them using the appropriate AI module, and display the resulting outputs.

The footwear detection module uses YOLOv8 and YOLO-World to identify and classify footwear objects. The virtual try-on component processes person and garment images using the selected try-on model and the required image-preprocessing procedures. The Streamlit interface provides access to the available functionalities and displays their outputs.

The development environment includes an AMD Ryzen 5 5500U processor and 32 GB of RAM. The system can use CPU-based execution for compatible models, although processing time depends on the model architecture, input resolution, and computational requirements.

The evaluation focuses on checking the functionality of the individual modules, the integration of the processing pipeline, and the presentation of outputs through the web interface. Quantitative performance values must be obtained from actual test runs before conclusions about model accuracy or efficiency can be established.

5.2 Performance of the Footwear Detection Module

The footwear detection module is designed to identify footwear objects and classify them into the categories supported by the selected model. YOLOv8 and YOLO-World form the foundation of this module.

The detection workflow processes an input image and returns the predicted object locations and class labels. Where available, confidence scores provide additional information about the model's predictions. These outputs allow users to inspect the detected footwear through the Streamlit interface.

The module supports the project's objective of automating footwear identification. However, successful detection in individual images does not establish the model's overall performance. A systematic evaluation using a separate test dataset is required to measure accuracy-related metrics, identify classification errors, and assess performance across different footwear categories.

The following aspects are relevant to evaluating the module:

- Identification of footwear objects in input images.
- Classification into the supported footwear categories.
- Localization of detected objects using bounding boxes.
- Consistency of predictions across images with different backgrounds and lighting conditions.
- Inference time when processing an image.

The actual detection outputs and measured evaluation metrics should be used to establish the final performance of this module.

5.3 Performance of the Virtual Try-On Module

The virtual try-on module is intended to generate a visual representation of clothing on a person image. It accepts the required person and garment images and processes them through the selected virtual try-on pipeline.

Image preprocessing, human parsing, and mask generation are relevant to preparing inputs for the try-on model. These operations help identify the regions required for garment visualization and support the image-synthesis process.

The resulting image can help users explore the visual appearance of clothing before making a purchasing decision. However, the quality of the output depends on the selected model, input-image quality, mask accuracy, and image alignment.

The virtual try-on module can be evaluated by examining:

- Whether the input person and garment images are processed correctly.
- Whether the selected garment is represented in the generated output.
- Whether the person's visible body regions and clothing boundaries are preserved appropriately.
- Whether image artefacts or alignment errors are present.
- Whether the model completes inference within an acceptable time.

The virtual try-on output represents a visual approximation and does not guarantee actual clothing fit, size, comfort, or physical appearance. Final conclusions about output quality should be based on the generated images and a defined evaluation procedure.

5.4 Integration into the Streamlit Application

The Streamlit application provides a common interface for accessing the implemented functionalities of ZcannerReality. It allows users to upload images, initiate the available processing operations, and view the corresponding results.

The application integrates the user interface with the footwear detection and virtual try-on workflows. The detection module returns information about identified footwear objects, while the virtual try-on module produces a generated image based on the supplied inputs.

The application-level evaluation focuses on the following aspects:

1. Input Handling: The interface should accept the required images and provide suitable feedback when inputs are missing or invalid.

2. Module Execution: The selected functionality should invoke the corresponding processing pipeline.

3. Result Display: Detection results and generated images should be presented in the interface.

4. Usability: The interface should allow users to access the available functionalities without directly interacting with the underlying model code.

5. Error Handling: The application should handle problems such as missing model weights, unsupported image formats, and inference failures appropriately.

The integrated interface provides the foundation for a unified product-visualization workflow. Additional application features can be introduced as the project develops.

5.5 Footwear Detection and Virtual Try-On: Functional Comparison

Table 5.1 summarizes the main differences between the two processing modules in ZcannerReality.

Aspect	Footwear Detection Module	Virtual Try-On Module
Primary purpose	Identify and classify footwear objects.	Generate a visual representation of clothing on a person image.
Input	Footwear image.	Person image and garment image.
Main technique	YOLOv8 / YOLO-World object detection.	Virtual try-on model and image preprocessing.
Processing output	Class labels, bounding boxes, and confidence scores where available.	Generated virtual try-on image.
Evaluation focus	Detection accuracy, localization, classification, and inference time.	Visual quality, garment representation, alignment, and inference time.
Main limitation	Predictions depend on model configuration, image quality, and supported categories.	Results depend on preprocessing, model capability, and input-image quality.

Table 5.1: Functional Comparison of ZcannerReality Modules

The two modules serve complementary purposes. Footwear detection identifies objects within an image, whereas virtual try-on generates a new visual representation using person and garment inputs. Their integration within one application provides a foundation for more interactive product exploration.

However, the two modules require different evaluation procedures. Detection metrics alone cannot establish the visual quality of try-on outputs, and visual inspection of generated images cannot establish the classification accuracy of the detection model.

5.6 Overall Analysis

The ZcannerReality architecture combines footwear detection, virtual try-on processing, and a web-based interface to address different aspects of online product visualization.

The footwear detection module supports automated recognition of footwear objects, reducing the need for users to identify products manually. The virtual try-on module aims to improve product visualization by generating an image showing selected clothing on a person. The Streamlit interface connects these functionalities and presents their outputs in an accessible form.

The integration provides several potential advantages:

- Automated Recognition: AI-based object detection supports footwear identification and classification.

- Personalized Visualization: Virtual try-on enables users to explore clothing on a person image rather than relying only on static product photographs.

- Unified Interaction: The web interface provides a common environment for accessing the available modules.

- Modular Development: Separate processing components allow individual models to be improved or replaced.

- Future Extensibility: The architecture provides a foundation for adding product categories and more comprehensive digital twin capabilities.

The effectiveness of these advantages depends on the reliability of the underlying models and their integration. Measured detection performance, try-on image quality, and inference time are necessary to determine how well the prototype meets its objectives.

Comprehensive digital twin generation and support for multiple product categories should not be considered completed capabilities unless they have been implemented and validated.

5.7 Quantitative Evaluation Metrics

Quantitative evaluation is necessary to assess the performance of the footwear detection model and the computational efficiency of the overall application. Table 5.2 provides a structure for recording measured results.

Metric	Purpose	Measured Value
Precision	Measures the proportion of predicted positive detections that are correct.	To be calculated from test results.
Recall	Measures the proportion of actual objects correctly detected.	To be calculated from test results.
mAP@0.5	Evaluates object detection performance at an intersection-over-union threshold of 0.5.	To be obtained from model evaluation.
mAP@0.5:0.95	Evaluates detection performance across multiple intersection-over-union thresholds.	To be obtained from model evaluation.
Inference Time	Measures the time required to process an input image.	To be measured during testing.
Try-On Processing Time	Measures the time required to generate a virtual try-on output.	To be measured during testing.

Table 5.2: Quantitative Evaluation Metrics

The detection metrics should be obtained by evaluating the model on a suitable test dataset. Inference time should be measured under a documented hardware and software configuration.

Virtual try-on quality may be evaluated through structured visual assessment or appropriate image-quality metrics, depending on the availability of reference outputs and the evaluation protocol. Detection metrics such as mAP are not appropriate substitutes for measuring the visual quality of generated try-on images.

No numerical performance values are asserted here because verified experiment logs and final evaluation measurements have not been supplied.

5.8 Limitations and Threats to Validity

The current evaluation has several limitations that must be considered when interpreting the prototype's capabilities.

- Detection Performance: The actual accuracy and generalization of the footwear detection model must be established through systematic evaluation on held-out test data.

- Virtual Try-On Quality: The generated output may contain alignment errors or visual artefacts, depending on the selected model and preprocessing pipeline.

- Computational Requirements: Model inference time and memory consumption may vary with input resolution, model complexity, and the available hardware.

- Category Coverage: Detection and try-on capabilities are limited to the categories and processing functions supported by the integrated models.

- Evaluation Scope: Individual successful outputs do not establish overall system reliability. Repeated tests across varied inputs are needed.

- Digital Twin Capabilities: Comprehensive digital twin generation remains a broader project objective and should not be treated as validated based solely on footwear detection or clothing try-on.

- Real-World Applicability: Virtual try-on provides a visual approximation. It does not independently establish product fit, comfort, or suitability for purchase.

These limitations identify areas for additional testing and improvement before the system can be considered a comprehensive virtual shopping solution.

5.9 Chapter Summary

This chapter described the evaluation approach for the ZcannerReality prototype, covering footwear detection, virtual try-on processing, Streamlit integration, functional comparison, quantitative metrics, and evaluation limitations.

The system architecture brings together AI-based footwear recognition and clothing visualization within a common web interface. Further testing is required to establish measured detection

performance, image-generation quality, and processing efficiency. The findings from systematic evaluation can guide future improvements to the detection models, virtual try-on pipeline, and application interface.

Chapter 6 discusses the potential applications of ZcannerReality, demonstrates its intended user workflows, and examines the opportunities for future development.

CHAPTER 6
ZCANNERREALITY APPLICATIONS AND DEMONSTRATION

6.1 Application Overview

The ZcannerReality prototype includes a virtual try-on application designed to support interactive clothing visualization. The application provides a graphical interface through which users can start the camera, select a garment, enable real-time try-on, and generate a high-quality CatV2TON result. The interface separates the live camera view from the virtual try-on output, allowing users to observe the input and generated result side by side.

As shown in the provided application screenshot, the interface includes controls for starting and stopping the camera, selecting a garment, enabling real-time virtual try-on, generating a high-quality CatV2TON result, displaying the generated result, and exiting the application. A status area displays information about the current operating mode, camera state, frame rate, rendering performance, tracking status, GPU memory usage, and selected garment.

The current interface provides the foundation for interactive virtual try-on. The displayed screenshot shows the application in a stopped state, with the camera turned off and no generated try-on result currently displayed.

6.2 Live Camera and Virtual Try-On Interface

Figure 6.1 presents the ZcannerReality virtual try-on interface. The application window is divided into two main panels: Live Camera and Current Try-On Output.

The Live Camera panel displays the camera feed when the camera is started. The Current Try-On Output panel is intended to display the generated virtual try-on result when real-time try-on is enabled or a CatV2TON result has been generated.

The interface provides the following controls:

Start Camera: Starts the camera to capture the user's live image.

Stop Camera: Stops the active camera feed.

Select Garment: Allows the user to choose a garment for visualization.

Enable Real-Time Try-On: Enables the real-time try-on functionality when the required components are available.

Generate High-Quality CatV2TON Result: Initiates the high-quality virtual try-on generation process.

Show CatV2TON Result: Displays the generated CatV2TON output.

Quit: Closes the application.

The status line provides information about the current mode, camera state, frame rate, rendering rate, tracking status, GPU memory usage, and selected garment. In the provided screenshot, the selected garment is `garment_transparent.png`, the camera is off, and the application is in the stopped state.

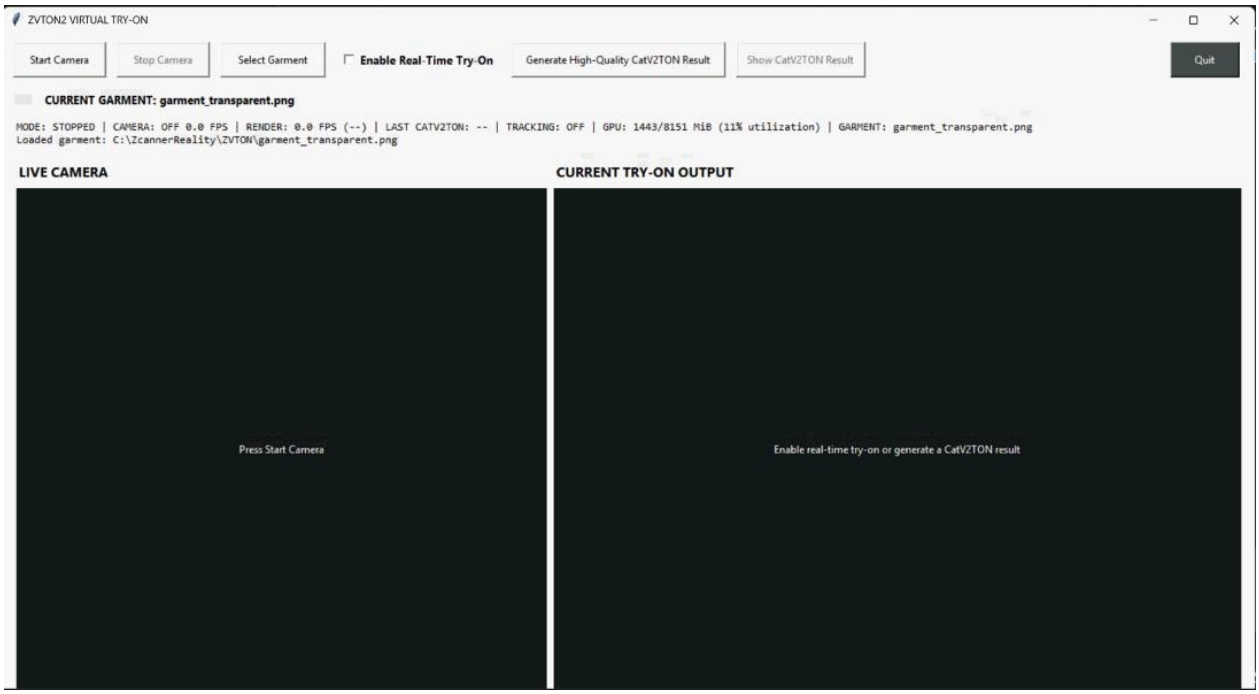
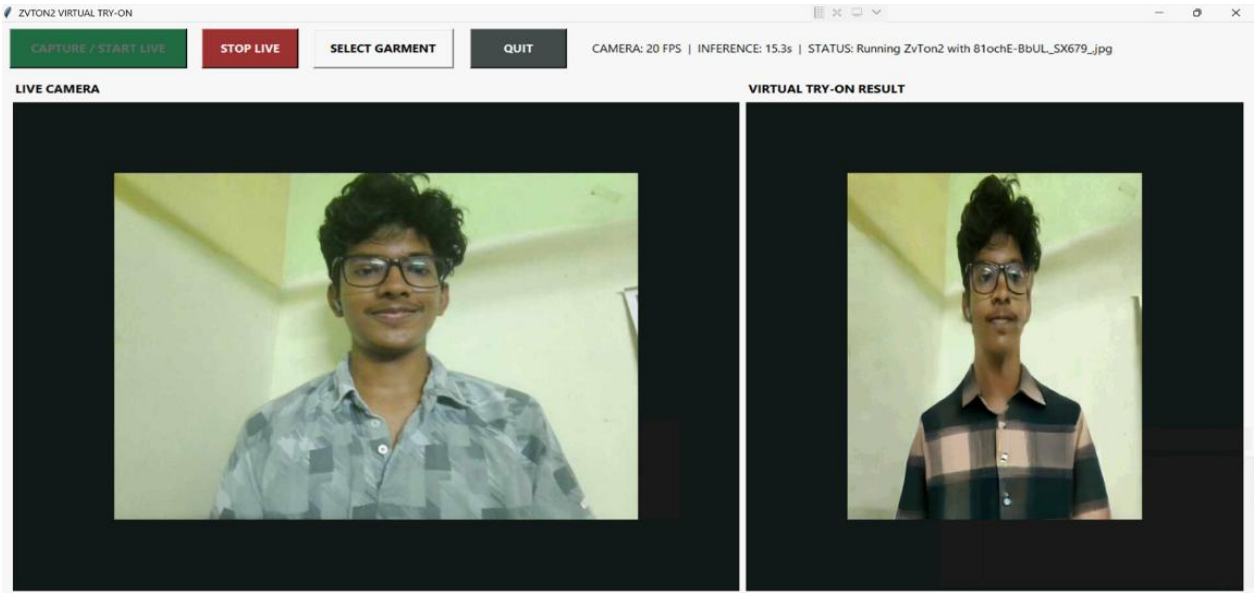


Figure 6.1: ZcannerReality Virtual Try-On Application Interface



6.3 Live Camera and Try-On Output Demonstration

The application is designed to present the input camera feed and virtual try-on result simultaneously. This arrangement allows users to compare the original person image with the generated visualization.

When the camera is started, the Live Camera panel is intended to show the captured image. After a garment is selected and the try-on process is executed, the Current Try-On Output panel is intended to display the generated result. The application also provides a separate control for generating a high-quality CatV2TON output.

In the supplied screenshot, both panels are currently dark, and the interface prompts the user to start the camera or enable real-time try-on or generate a CatV2TON result. Therefore, the screenshot demonstrates the application interface and its available controls, rather than a completed virtual try-on result.

The expected demonstration workflow is as follows:

Open the ZcannerReality virtual try-on application.

Start the camera to capture the person image.

Select the desired garment using the garment-selection control.

Enable real-time try-on or generate a high-quality CatV2TON result.

Display and inspect the generated output in the Current Try-On Output panel.

The actual quality and responsiveness of the generated result should be assessed after successful execution of the try-on pipeline.

6.4 System Status and Performance Information

The application includes a status area that reports operational information about the virtual try-on process. This information can help developers understand the state of the application and identify potential performance issues.

Status Field	Purpose
Mode	Indicates the current operating state of the application.
Camera	Shows whether the camera is active or inactive.
FPS	Reports the camera or processing frame rate when available.
Render FPS	Indicates the rendering rate of the try-on output when available.
Last CatV2TON	Provides status information about the most recent CatV2TON result.
Tracking	Indicates the tracking state of the application.
GPU Memory	Displays GPU memory information when supported by the execution environment.
Garment	Shows the filename of the currently selected garment.

Table 6.1: Virtual Try-On Application Status Indicators

These indicators provide useful diagnostic information during development. However, the values shown in the status line should be interpreted according to the application's actual execution state. A displayed GPU memory value does not, by itself, establish the quality or speed of virtual try-on generation.

6.5 Potential Applications

ZcannerReality has potential applications in several areas of digital retail and product visualization.

Online Fashion Retail: Virtual try-on can help customers visualize clothing using a person image before purchasing.

Footwear Product Recognition: AI-based detection can identify footwear categories in images.

Interactive Shopping Platforms: A unified interface can provide customers with access to product detection and visualization features.

Digital Product Demonstration: Retailers can use generated visualizations to make online product exploration more interactive.

Personalized Shopping Experiences: Future enhancements can support broader product categories and more personalized visual representations.

These are potential application areas. Their practical effectiveness depends on the capabilities of the implemented modules and further testing.

6.6 Chapter Summary

This chapter presented the ZcannerReality virtual try-on application and its interface, including camera controls, garment selection, real-time try-on controls, CatV2TON result generation, and status indicators. The supplied screenshot demonstrates the application's layout and available controls while the camera and output are inactive.

The application provides a foundation for interactive clothing visualization. Further demonstrations using actual generated try-on images and measured processing performance will help establish the quality and usability of the completed workflow.

CHAPTER 7
CONCLUSION AND FUTURE WORK

7.1 Conclusion

ZcannerReality demonstrates the potential of combining computer vision, deep learning, and virtual try-on technology to improve the online shopping experience. Traditional e-commerce platforms primarily rely on static product images and descriptions, which may not provide customers with sufficient opportunities to visualize products before purchasing them. By integrating AI-based footwear detection, virtual try-on processing, and a Streamlit-based interface, the project establishes a foundation for more interactive and personalized product visualization.

The current prototype focuses on identifying and classifying footwear using YOLOv8 and YOLO-World, integrating a clothing virtual try-on workflow, and providing an interface for users to interact with the available functionalities. Image preprocessing, human parsing, and mask generation support the preparation of inputs for virtual try-on processing. Together, these components demonstrate an approach to bringing product recognition and visual try-on capabilities into a common application. The broader objective is to develop a more comprehensive digital shopping experience that helps customers explore products visually and make more informed purchasing decisions.

7.2 Achievement of Project Objectives

The project objectives have been addressed through the development of the footwear detection module, the integration of virtual try-on functionality, and the creation of a Streamlit-based interface. Footwear detection has been implemented using YOLOv8 and YOLO-World, while the virtual try-on component provides a framework for generating clothing visualizations using person and garment images. Image preprocessing and human parsing support the try-on workflow, and the application interface brings the available functionalities together. Comprehensive digital twin generation, expanded multi-category visualization, and additional real-time interaction capabilities remain areas for further development. Quantitative model performance must be established through systematic testing and documented evaluation.

7.3 Main Contributions

The principal contribution of ZcannerReality is the development of a prototype that integrates AI-based footwear detection and virtual try-on functionality within a common application. The use of YOLOv8 and YOLO-World provides an automated approach to footwear identification and classification, while the virtual try-on pipeline supports clothing visualization on a person image. The project also incorporates image preprocessing techniques relevant to preparing inputs for try-on generation and uses Streamlit to provide an accessible interface for interacting with the implemented modules. The resulting architecture provides a foundation for extending the application towards broader product visualization and more comprehensive digital twin capabilities.

7.4 Limitations

The current prototype has limitations that must be considered when interpreting its capabilities. Footwear detection performance depends on the model configuration, dataset quality, and variation in input images. Virtual try-on quality depends on the selected model, image

preprocessing, mask generation, and alignment between the person and garment images. The computational requirements of deep-learning models may also affect inference time on systems without dedicated GPU acceleration. In addition, the current implementation does not establish comprehensive digital twin generation or complete virtual try-on support for all product categories. Further testing is required to measure detection performance, evaluate generated image quality, and assess the efficiency and reliability of the integrated application.

7.5 Future Enhancements

Future development will focus on extending the system beyond its current footwear detection and clothing virtual try-on capabilities. Additional product categories, including trousers, accessories, and other fashion items, may be incorporated through suitable models and preprocessing pipelines. Improvements to human parsing, mask generation, and image alignment could help produce more consistent virtual try-on outputs. Real-time camera interaction and improved inference efficiency may also make the application more responsive and convenient to use.

Further work will explore more comprehensive digital twin generation, allowing customers to interact with personalized representations of products and potentially compare multiple product options within a unified environment. Additional evaluation using diverse images and documented performance measurements will be important for assessing the system's effectiveness. These enhancements aim to develop ZcannerReality into a more flexible and personalized AI-assisted shopping platform.

7.6 Final Summary

In conclusion, ZcannerReality establishes a foundation for improving online product visualization through the integration of footwear detection, virtual try-on processing, and a web-based user interface. The current prototype brings together computer vision and deep-learning techniques to support footwear identification and clothing visualization. Future improvements in model performance, image processing, multi-category support, and digital twin generation can extend these capabilities towards a more comprehensive virtual shopping experience, helping customers explore products in a more interactive and personalized way.

CHAPTER 8
REFERENCES

[1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[2] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal Speed and Accuracy of Object Detection," arXiv preprint arXiv:2004.10934, 2020.

[3] G. Jocher et al., "Ultralytics YOLOv8," Ultralytics, 2023. Available: https://github.com/ultralytics/ultralytics

[4] T. Cheng et al., "YOLO-World: Real-Time Open-Vocabulary Object Detection," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Cheng_YOLO-World_Real-Time_Open-Vocabulary_Object_Detection_CVPR_2024_paper.html

[5] S. Choi et al., "VITON-HD: High-Resolution Virtual Try-On via Misalignment-Aware Normalization," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021. Available: https://openaccess.thecvf.com/content/CVPR2021/html/Choi_VITON-HD_High-Resolution_Virtual_Try-On_via_Misalignment-Aware_Normalization_CVPR_2021_paper.html

[6] E. Xie et al., "SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers," in Advances in Neural Information Processing Systems (NeurIPS), 2021. Available: https://papers.nips.cc/paper/2021/hash/64f1f27bf1b4ec22924fd0acb550c235-Abstract.html

[7] A. Vaswani et al., "Attention Is All You Need," in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.

[8] A. Dosovitskiy et al., "An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale," in International Conference on Learning Representations (ICLR), 2021.

CHAPTER 9
APPENDIX

9.1 Prototype Artefacts

The principal artefacts associated with the ZcannerReality project are listed in Table 9.1. These artefacts include the footwear detection models, virtual try-on components, datasets, and web application used in the development of the prototype.

Artefact	Type	Description
YOLOv8 model	Deep-learning model	Used for detecting and classifying footwear objects in images.
YOLO-World model	Open-vocabulary object detection model	Supports flexible object detection using text-based category descriptions.
Footwear Detection Dataset	Image dataset	Contains annotated footwear images for model training, validation, and testing.
VITON-HD	Virtual try-on dataset and research implementation	Provides resources for high-resolution clothing virtual try-on research.
Virtual Try-On Module	Python-based processing module	Processes person and garment images to generate clothing visualization using the integrated try-on model.
Image Preprocessing Module	Image-processing component	Prepares person and garment images and supports human parsing and mask generation where required.
Streamlit Application	Web application	Provides the user interface for accessing footwear detection and virtual try-on functionality.
ZVTON2 Virtual Try-On Interface	Graphical user interface	Provides camera controls, garment selection, real-time try-on controls, and CatV2TON result-generation options, as shown in the application screenshot.

Table 9.1: ZcannerReality Prototype Artefacts

9.2 Information to Be Completed by the Authors

The following information should be verified and completed before final submission of the project report. These details should reflect the actual implementation and institutional requirements.

Item	Information Required
Institutional Details	Institution name, department name, degree, and academic year.
Student Details	Student names and register numbers.
Guide Details	Guide's name, designation, and department.

Dataset Information	Dataset sizes, class distribution, and train-validation-test split.
Model Configuration	Exact YOLOv8 and YOLO-World model versions, weights, and training parameters.
Virtual Try-On Backend	Exact repository, model name, model weights, and version used in the final implementation.
Evaluation Results	Actual detection metrics, inference time, and virtual try-on evaluation results, where measured.
Application Evidence	Actual screenshots of footwear detection, generated virtual try-on outputs, and the application interface.
Software Environment	Python version, dependency versions, and execution configuration.
Implementation Status	Confirmation of which preprocessing, human parsing, and virtual try-on components are fully integrated.

This appendix provides a record of the principal software artefacts and the information required to document the project accurately. All model-performance values, implementation details, and experimental findings should be included only when supported by the actual source code, execution results, or experiment logs.

CHAPTER 10
SOURCE CODE DOCUMENTATION

10.1 Overview of the Implementation Codebase

The ZcannerReality project is designed to identify footwear using deep learning and provide a virtual try-on interface. The implementation combines object detection, dataset configuration, trained model weights, image processing, and a user interface for interacting with the system.

The footwear detection component uses YOLO-based object detection, with YOLOv8 and YOLO-World explored during development. The dataset is organized into training, validation, and testing folders, and its class names are specified in a `data.yaml` configuration file. The virtual try-on component is intended to process user-provided images through a compatible try-on backend.

The major implementation components are:

Footwear detection: Identifies footwear categories in input images.

Dataset configuration: Defines dataset locations and detection classes.

Model training and inference: Uses trained model weights for prediction.

Image processing: Prepares images for detection and try-on.

Virtual try-on integration: Connects the interface to a suitable try-on model.

User interface: Allows users to provide images and view available outputs.

The following directory structure is an illustrative layout. The exact filenames and arrangement must be verified against the final project folder.

```
ZcannerReality/
├─── data.yaml
├─── train.py
├─── detect.py
├─── best.pt
├─── datasets/
│    ├─── train/
│    ├─── valid/
│    └─── test/
├─── virtual_tryon/
│    ├─── preprocessing.py
│    ├─── masks.py
│    └─── inference.py
├─── app.py
└─── requirements.txt
```

10.2 Footwear Detection Using YOLO

The footwear detection module uses a YOLO-based model to identify footwear objects in an input image. The model predicts object locations using bounding boxes and assigns the detected objects to the supported footwear classes.

The dataset used during development contains the following classes: Boots, Heels, Sandals, Shoes, Slipers, and Sneakers. The spelling `Slipers` is retained here as it appears in the supplied dataset description; it should be corrected consistently if the dataset configuration uses `Slippers`.

The general inference process is:

Load the trained YOLO model.

Read the input image.

Run object detection.

Retrieve predicted classes, bounding boxes, and confidence scores

Display the detected footwear and associated labels

The following is an illustrative example of the detection process, not a verified copy of the project's exact source code.

```
from ultralytics import YOLO



# Load the trained footwear detection model

model = YOLO("best.pt")



# Perform inference on an input image

results = model("input.jpg")



# Display the detection results

for result in results:

    result.show()
```

The model file `best.pt` is an example filename. The actual path must match the trained weights saved in the ZcannerReality project.

10.3 Dataset Configuration and Model Training

The footwear dataset is organized into training, validation, and testing subsets. The training subset is used to learn object features, the validation subset supports model monitoring during training, and the testing subset can be used for final evaluation.

The `data.yaml` configuration file specifies the dataset paths and the number and names of the classes.

An illustrative configuration is shown below. The paths and class names must match the actual dataset configuration.

```
train: path/to/dataset/train/images
val: path/to/dataset/valid/images
test: path/to/dataset/test/images

nc: 6

names:
  - Boots
  - Heels
  - Sandals
  - Shoes
  - Slipers
  - Sneakers
```

The YOLO training process can be implemented using the Ultralytics framework. The following example demonstrates the general training command:

```
from ultralytics import YOLO



# Load a YOLO model

model = YOLO("yolov8s.pt")



# Train using the configured footwear dataset

model.train(

    data="data.yaml",

    epochs=50,

    imgsz=640,

    batch=8

)
```

The model variant, training duration, batch size, and image dimensions must reflect the actual experiment. These example settings should not be reported as completed training results unless they match the recorded run.

10.4 Virtual Try-On Processing and Application Interface

The virtual try-on module is intended to combine an input image with a selected product or garment image to generate a try-on result. Depending on the selected backend, the processing

pipeline may include image resizing, human parsing, mask generation, model inference, and output rendering.

The preprocessing stage prepares the input images for the model. Human parsing can help identify relevant body regions, while masks can specify the areas that should be modified during try-on generation. The processed inputs are then passed to the selected virtual try-on backend.

The general processing sequence is:

Accept the user's input image.

Load the selected product or garment image.

Preprocess the input images.

Generate required masks or parsing information when supported by the backend.

Run virtual try-on inference.

Display the generated output if inference completes successfully.

An illustrative interface-level workflow is shown below:

```
def run_virtual_tryon(person_image, garment_image, tryon_backend):
    """
    Illustrative workflow.
    The actual preprocessing and inference calls depend
    on the selected virtual try-on implementation.
    """
    prepared_person = preprocess_person(person_image)
    prepared_garment = preprocess_garment(garment_image)

    result = tryon_backend(
        prepared_person,
        prepared_garment
    )

    return result
```

The functions in this example describe the workflow and are not verified function names from the existing codebase. The actual backend must also support the intended product category; a clothing try-on model cannot automatically be assumed to support footwear.

The application interface connects the detection and try-on components where integration is available. It should present the input image, detection information, selected item, processing status, and generated output. If model inference fails or no output is generated, the interface should report that condition rather than display an assumed result.

CHAPTER 11
CONFERENCE APPLICATION

ZcannerReality: An AI-Assisted 3D Virtual Try-On and Augmented Reality Framework for Personalized Fashion Visualization

Jaison J[1], Gnaneshwaran S[2], Karthikeyan P[3], Dr.Ulaga Priya K[4]
[123]UG Scholars , [4]Associate Professor
Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)
VELS Institute of Science, Technology & Advanced Studies, Chennai, India
jj3632056@gmail.com , 24619112@vistas.ac.in , 24619147@vistas.ac.in , upriya.se@vistas.ac.in

Abstract

Online fashion shopping offers customers many clothing options, but it is often difficult to imagine how an outfit will look on their own body before purchasing it. ZcannerReality is a proposed AI-powered virtual try-on system designed to make online shopping more interactive by helping users visualize clothing on a digital representation of their body. The system takes a user's image and processes it using computer vision techniques to identify body parts, estimate posture, and reconstruct body shape. This information helps position the selected garment more naturally on the digital model. By combining Artificial Intelligence (AI) with Augmented Reality (AR), the system aims to provide a more realistic preview than simply placing a clothing image over a photograph. Technologies such as OpenCV, PyTorch, MediaPipe, and suitable 3D and AR tools can support its development. The framework also considers practical challenges, including varying lighting conditions, overlapping body parts, changes in posture, and the processing limitations of mobile devices. Its effectiveness will be evaluated through future experiments that examine garment alignment, response time, performance under different conditions, and perceived realism. Ultimately, ZcannerReality aims to simplify online clothing selection and help customers make more informed purchasing decisions.

Keywords: Virtual Try-On, Augmented Reality, 3D Body Reconstruction, Human Pose Estimation, Digital Twin, Garment Fitting, Fashion Technology

Institute of Computer Science and Engineering, SIMATS Engineering, Chennai.
ISBN No.: 978-93-344-7370-4

Conference: 2026 ICAI-CEME SIMATS Engineering

ISBN No:978-93344-7370-4